Eclipson eVTOL Drone Build for Wildlife Monitoring and Anti-Poaching Patrol

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I. Executive Summary

This Report covers the construction and analysis of a fixed-wing eVTOL platform designed by Eclipson. The aim of the project was to build a long-range and efficient low-noise aircraft, capable of conducting patrols of a reserve without the need for a runway.

The design features a tail-sitter VTOL configuration for vertical take-off and fixed-wing cruise,

The drone's airframe was fabricated using 3D printers, and all other parts and equipment were supplied by Durham University. Both Mission Planner and LUA code were used to integrate the software into the platform, providing autonomous capabilities. Despite correctly adhering to the set-up guide, published by Eclipson, multiple structural limitations stopped us from achieving the goal of transitioning to fix-wing cruise flight.

Specifically, weaknesses between the wings and fuselage that resulted in difficulty to land without the drone suffering significant damage. Large time delays came as a consequence of the entire wing being re-printed. This made it hard to troubleshoot calibration and sensitivity issues in the given time frame.

During the test flight we encountered minor errors surrounding hardware, GPS calibration and electromagnetic interference. Alone, these were reparable issues, but due to a lack of repairability, the iterative troubleshooting process took longer than expected.

Recommendations were made for improvements to both the design, and the build process. These included suggestions to improve the strength, feasibility and performance of the drone.

II. Business Case Background & Motivation

It is becoming increasingly difficult for conservation teams to monitor vast wildlife reserves to prevent threats like poaching, encroachment, and habitat loss. Traditional methods like ranger patrols and manned aircraft (helicopters) are expensive, time-consuming and limited in their ability to access all of the reserve. While a quad-copter setup offers some degree of flexibility, short flight times and a short range make them less viable for large scale applications.

The utilisation of fixed-wing, eVTOL drones offer a seemingly promising alternative. They combine the ease of vertical take-off that a drone provides with the long-range, low-noise and efficient cruise of a fixed-wing vehicle. With no need for a runway and less disturbance created, the Eclipson eVTOL has the potential to be ideal for patrol missions in remote or sensitive environments.

The platform could follow piloted or pre-programmed flight paths in an effort to gather geo-tagged imagery for vast areas with little to no disturbance to wildlife. Applications include species tracking, environmental monitoring and anti-poaching patrols in real-time. The setup would allow for the integration of optical or thermal payloads enhancing the scale and responsiveness of conversational operations while decreasing costs and risks to humans.

III. Problem Statement

The aim of this report is to explore the development of an easy to launch, efficient, low-noise and long range aerial platform to aid with the surveying of conservation areas with minimal user input. The purpose of this is to help wildlife organisations that lack the resources to conduct regular long-distance patrols across their reserves.

IV. Other Applications of eVTOL Drones

eVTOL drones are particularly useful for applications requiring a stationary or slow-moving

hover, separated by large distances. Specific examples include sampling water along the length of a river, package delivery, or powerline inspection. Companies such as Amazon have started rolling out eVTOL drones such as the mk23 and mk27-2 which have been approved by the FAA to fly outside of visual range [1].

V. Technical Build

A. Airframe Assembly

The 3D printed parts were assembled using cyanoacrylate (superglue) following an instructional video provided by Eclipson. The fuselage, consisting of two sections and a carbon rod, includes a main central compartment housing the battery and other hardware such as the flight controller and GPS. The wings were assembled of multiple parts attached using glue as prescribed, which included the encasing of the elevons, actuated by servo motors screwed into the wings. The vertical stabiliser was attached to aid with stability and takeoff. This included a carbon rod from the tip of the top piece to the bottom piece for structural integrity, essential for landing, when all the weight of the eVTOL is held by the 4 wings as shown in Figure 1.



Fig. 1: Completed Assembly

B. Hardware Integration and Power Distribution

The electrical architecture allowed for the reliable delivery of power and communication across all subsystems. The system was centred around a power distribution board (PDB), which served as a hub for electrical routing. The LiPo battery was connected to the PDB through a splitter, shown in Figure 2, supplying DC voltage to all downstream components. Two electronic speed controllers (ESCs) were connected to the PDB, responsible for regulating power to each motor and therefore the speed of the propeller.

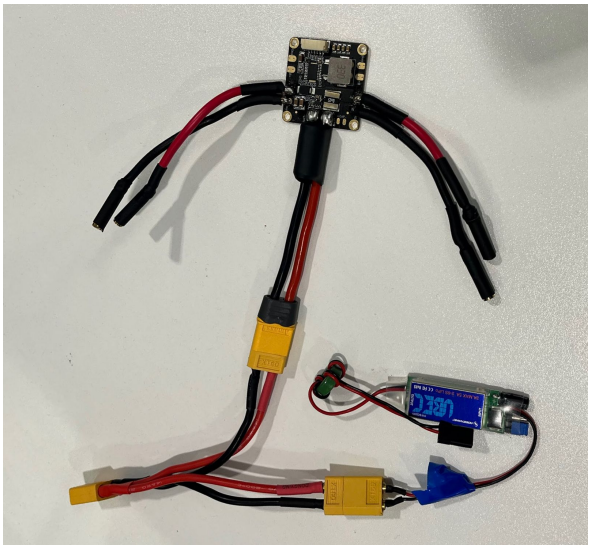


Fig. 2: PDB (top), Splitter, and Buck Converter

The flight controller also received regulated power via a connection to the PDB and functioned as the aircraft's central processing unit (CPU). The flight controller sends throttle signals through PWM connections to the ESCs, making the necessary adjustments to the motor speed. This enables the coordinated control of the motor during takeoff, transition, cruise flight and landing. The servos controlling the elevons required power and a PWM signal to control the angle. PWM signals are supplied to the motors and servos by the flight controller's pins as shown in Figure 3.

With the servos requiring power, the splitter powers the PWM pins' rail using a buck down converter connected to PWM pin 8. Other peripheral systems were also connected to the flight controller. A GPS module concerned with collecting positional data and navigation is interfaced via a Universal Asynchronous Receiver-Transmitter. The telemetry transmitter is also linked to the controller which exchanges real-time data with ground control. The completed hardware assembly is seen in figure 4.

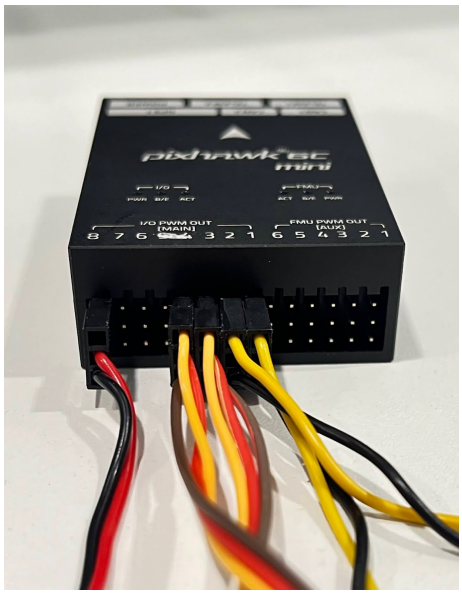


Fig. 3: PWM pins, red live, brown/black ground, yellow signal

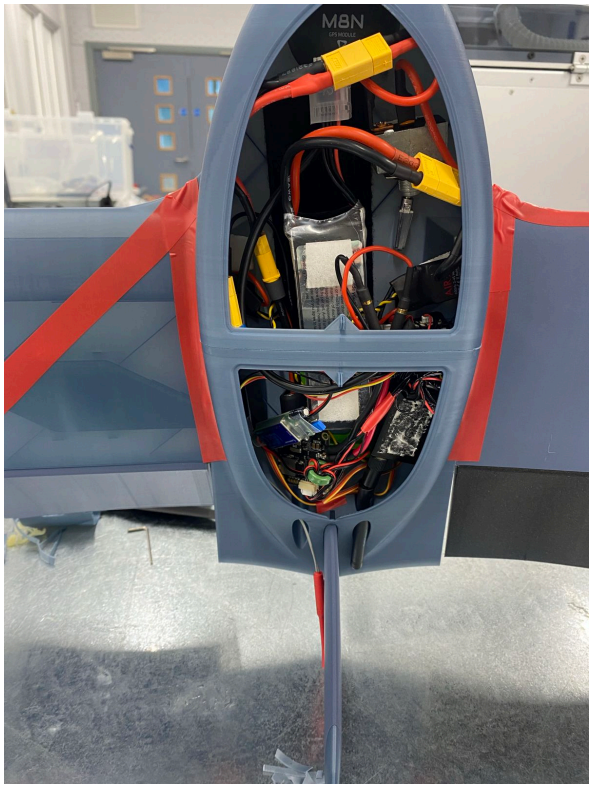


Fig. 4: Complete hardware assembly

C. Flight Controller

The flight controller used in the eVTOL is a Holybro Pixhawk 6C Mini. Connected to the flight controller are the RC receiver, telemetry, GPS, motor ESC, and PDM.

D. Ground Control

The eVTOL drone was controlled using Mission Planner, an open-source software used for ground control. It is compatible with the flight controller ArduPilot flight stack. It allowed for complete mission planning, tuning and the integration of the flight modes associated with VTOL. Patrol routes were programmed using GPS waypoints to measure the drone's stability and energy usage over repeatable routes. Other key executions like take-off, landing and VTOL transitions could also be configured and tested using the software. Mission planner would have been vital to validating the drone's autonomous performance, allowing for consistency across test flights and providing a medium for data collection.

VI. Simulation and Programming

VII. Flight Testing

A. Preflight

Prior to the first flight, the drone was tested in the manual flight mode on the ground. This allowed the flaps to be controlled to ensure the correct trim settings and that RC inputs were correctly interpreted. Additionally, logging was enabled to capture all data from the flight controller for analysis.

Having confirmed successful flap operation, the drone was able to be tested during the first flight.

B. First Flight

During the first flight, the drone was unable to take off from the ground without flipping over. After reviewing footage of the flight, it was determined that the cause of failure was the metal wire feet shown in Figure 5. Using a grassy field with patches of moss caused the feet to become stuck in the ground, resulting in the drone pivoting around the stuck feet. Luckily, the drone was undamaged.

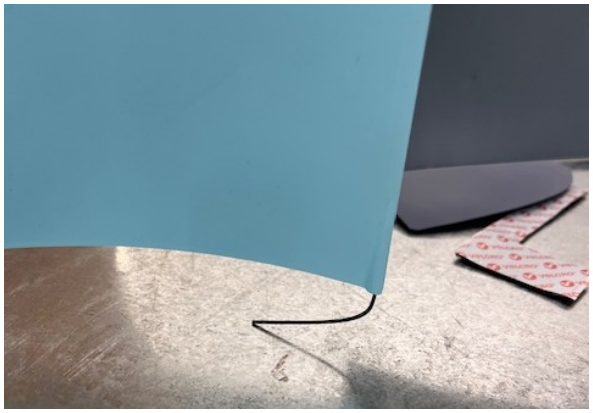


Fig. 5: Wire Feet

C. Second Flight

The second flight aimed to achieve a successful hover. To prevent the issues with the feet getting stuck, the drone took off from a metal plate. However, upon increasing the throttle, the drone flipped backwards and to the side, causing the wings to be damaged beyond repair as shown in Figure 6.



Fig. 6: Wing Damage

D. Crash Analysis

Before the second flight, Mission Planner's HUD was showing that the drone was slightly off level while sitting on its feet; this was corrected manually by adjusting parameters in Mission Planner. Following the second attempted flight, the logs were downloaded from the flight controller. However, despite setting the logging bitmask to capture all data, the desired pitch was not captured, making it difficult to rule out this being the error.

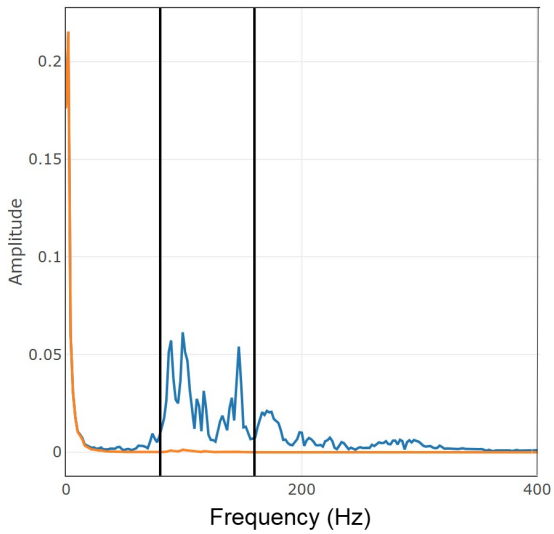


Fig. 7: Comparision of vibrations before (blue) and after (orange) FFT notch filters

VIII. Proposed Testing

Although a stable basis for flight testing was never achieved, a plan for testing and evaluation of the platform was put in place.

A. Hover Tuning

The initial step for achieving stable flight involved manually setting PID values, after a short initial flight, in particular the P and I gains related to the throttle controller, using the recommended defaults suggested in the ArduPilot parameter tree. Just before tuning, a Fast Fourier Transform (FFT) analysis would have been carried out to isolate vibrational noise from the motor and airframe oscillations. Notch filters (black lines in Figure 7) in the flight controller would have been configured from this data. Removing high frequency noise from sensor data would allow the flight controller to more accurately interpret the real-time movement of the drone.

Once noise filtering was integrated, the Quik Tune function would be activated to stabilise the hover. Quik Tune is an automatic tool used to

adjust control gains to stabilise pitch, roll and yaw. Following this, AutoTune would be used to refine the response characteristics of the drone. AutoTune systematically excites each axis of motion and observes the response, altering the stabilisation values to further improve stability.

B. Transition Testing

The performance of vertical take-off and hover to cruise transitions would be evaluated through test flights. Transition time and altitude gain/loss would be measured using data logged using Mission Planner. The smoothness and consistency of the transition could be measure by looking at the pitch angle and throttle response from flight logs. A successful transition would be predictable, with minimal altitude loss.

C. Efficiency Testing

The Eclipson website promises triple the flight distance for the same amount of energy as a standard quad-copter [2]. To test this claim a pre-programmed flight path would have been flown on a fully charged battery. Three times in pure hover mode and three times using transitions and fixed-wing flight. The distance travelled before the battery fail-safe initiates would have been recorded. The results would be averaged and compared to quantify efficiency gains.

D. Repeatability Testing

To test the reliability of the drone's navigation and autonomous flight capabilities, the final experiment involved executing multiple of the same predefined GPS waypoint missions. Flight paths would be compared to each other as well as with the programmed route to assess positional accuracy. This would have utilised the telemetry logs and deviation could have been quantified as RMS error. Stability and speed across all the missions could also be compared, highlighting how much impact varying external conditions have on these parameters.

IX. Lua Scripting

The initial objective of the Lua scripting was to achieve practical implementation on the eVTOL drone by the end of the week. Scripts loaded onto the eVTOL would have provided comprehensive control and flexibility over its operation. It would have also enabled the drone to function entirely autonomously, performing pre-arm checks, initiating take-off in hover mode, transitioning to fixed-wing flight, and systematically executing complex tasks such as conducting surveillance for poacher detection in wildlife reserves. Consequently, significant attention was devoted to the development of the control programs intended for the eVTOL. However, as the week progressed, it became apparent that the physical platform of the eVTOL was not yet sufficiently developed to execute the complex tasks defined in the scripts. As a result, the programs were tested exclusively within the Mission Planner simulation environment. Nevertheless, this process highlighted several key considerations for the future practical implementation of Lua scripts on a real eVTOL platform.

A. Early Development

After enabling scripting within Mission Planner, it was necessary to set the `Q_ENABLE` parameter to a value of `1`, as the VTOL was simulated using the ArduPlane firmware. Enabling this parameter activated a collection of additional configuration options specific to tailsitter aircraft - such as tailsitting behaviour, minimum tail transition angles, and frame class definitions - which are not available in ArduCopter.

The initial scripting task was to enable the VTOL to perform automated pre-arm checks. Once these checks were successfully completed, the VTOL would proceed to arm the throttle and autonomously take off to a predefined altitude. An illustration of the scripting methodology used to achieve this functionality is presented in Figure 8.

After experimenting with different jobs in order to develop a deeper familiarity with Lua scripting - for example, going around in circles as seen in Figure 9 - a comprehensive flow diagram was proposed in order to execute the VTOLs commercial intention, monitoring and anti-poaching patrol.

The most involved stage of this was configuring the transition from Hover mode to fixed-wing. This was initiated by the code in Figure 10. The particular parameters for transitioning into FBWA - tailsit-angle and angular rates - were fine tuned after experimenting with various ratios. Their effect is illustrated in Figure 11.

```
-- ------------------------------------------------------
-- Stage 1: Climb to TAKEOFF_ALT, record home & heading
-- ------------------------------------------------------
elseif stage == 1 then
  local home = ahrs and ahrs:get_home()
  local pos  = ahrs and ahrs:get_location()
  if home and pos then
    local d     = home:get_distance_NED(pos)
    local alt_m = -d:z()
    gcs:send_text(6, "Climbing: " .. math.floor(alt_m) .. " m")
    if alt_m >= (TAKEOFF_ALT - 1) then
      -- record home & heading
      home_loc  = home
      local yaw = ahrs:get_yaw()
      yaw_cos   = math.cos(yaw)
      yaw_sin   = math.sin(yaw)
      gcs:send_text(0, "Reached " .. TAKEOFF_ALT .. " m; switching to GUIDED")
      vehicle:set_mode(MODE_GUIDED)
      stage = 2
      retry_count = 0
    end
  else
    retry_count = retry_count + 1
    if retry_count % 10 == 0 then
      gcs:send_text(4, "Waiting for GPS/home…")
    end
  end
```

Fig. 8: Extract from Lua script for taking-off.

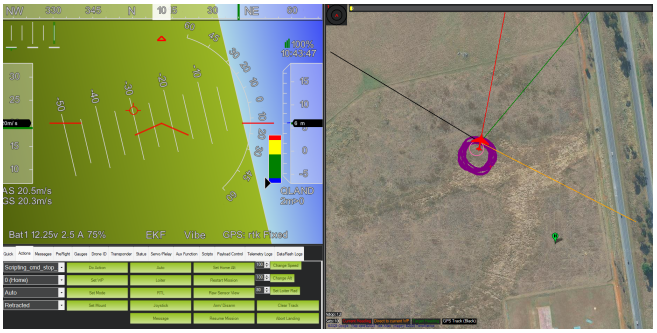


Fig. 9: Mission Planner simulation of a repeatable circular path.

```
elseif stage == 1 then
  local pos = ahrs and ahrs:get_location()
  local home_pos = ahrs and ahrs:get_home()
  if pos and home_pos then
    local d = home_pos:get_distance_NED(pos)
    local alt = -d:z()
    gcs:send_text(6, "Climbing: " .. math.floor(alt) .. "m")
    if alt >= TAKEOFF_ALT - 1 then
      gcs:send_text(0, "Reached altitude. Transitioning to FBWA")
      vehicle:set_mode(FBWA_MODE)
      stage = 2
      retry = 0
    end
  end
```

Fig. 10: Extract from Lua script for transitioning to FBWA.

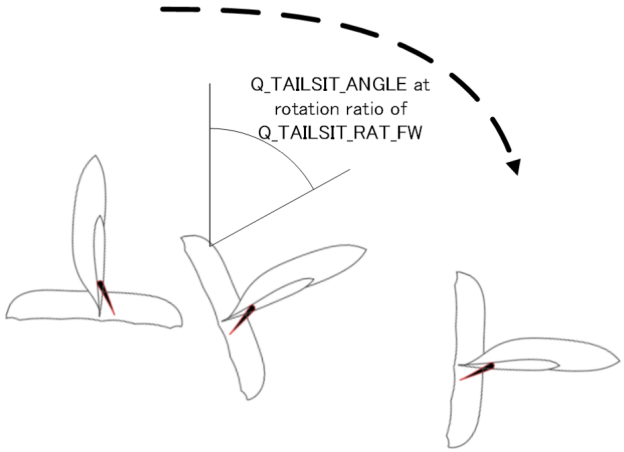


Fig. 11: Illustration of transition with relevant parameters.

B. Further Development

Once the VTOL had successfully transitioned to fixed-wing mode, the programming of its route-following behaviour was relatively straightforward. After parametrising four waypoints - positioned for this case study within Kruger National Park in South Africa - a script was developed to control the number of lawnmower-style passes using a simple loop counter. Increasing the number of passes between the waypoint pairs allowed for more rigorous area coverage, though at the cost of additional time and energy consumption.

A key limitation encountered during simulation was the inability to perform quick-tuning procedures to obtain suitable PID values. While the simulated VTOL executed its flight path without issue - due to pre-tuned settings within the simulation environment - practical deployment would require the development of a quick-tuning script to mitigate undesired oscillations and ensure flight stability.

Upon completing the prescribed number of lawnmower passes, the VTOL would autonomously return to its home location, thereby concluding the mission.

C. Future Development

As previously noted, it became apparent early in the project that the Lua script would not be practically implemented on the VTOL within the available time frame. Nonetheless, this limitation

presented an opportunity to reflect on the additional developments required to transform the VTOL into a commercially viable system. One such consideration involved enabling the real-time transmission of camera data back to the user. While Lua scripting is well-suited for triggering camera activation, real-time video transmission would necessitate the integration of an analogue video transmitter (VTX) or a digital link, such as the DJI FPV system.

To extend the system's capabilities further, a lightweight program written in Python using the OpenCV library could be developed to detect poachers based on movement patterns and body outlines captured in the live video feed. Upon autonomous identification of a suspected poacher, relevant anti-poaching authorities could be alerted with precise location data, allowing for timely and targeted intervention. Automating this process would enhance operational reliability and eliminate the need for continuous human monitoring of the transmitted video feed. Such a system has the potential to deliver a significant and lasting impact in the field of anti-poaching surveillance and intervention.

X. Advantages over Other Systems

The primary advantage for choosing general VTOL systems over quad-copter systems comes down to energy requirements. This can be attributed to the aerodynamic lift on the wings during forward flight - something that is absent in standard quad-copter systems altogether. This substantially reduces the power load on their motors compared to quad-copters, which must continuously use significant energy to generate lift through propeller thrust throughout the entire mission. This superior aerodynamic efficiency directly translates into significantly longer flight times and more efficient energy utilization - often in the region of 2-3 times the flight time of quad-copters. This offers the drone a greater range to operate in lending itself nicely to operating in wide expanses of savannah and nature reserves. With greater efficiency and fewer rotors, the eVTOL is also quieter than a conventional copter during forward flight.

Another advantage of the VTOL is the take-off and landing flexibility compared to fixed-wind aircraft. The VTOL does not require a runway, catapult or any other recovery system. This therefore allows it to work in confined spaces such as dense urban environments, or challenging terrains where ground infrastructure may be limited such as in sub-Saharan Africa in its role identifying poachers.

The final advantage is the payload capacity over a large range. Due to VTOL's more robust design, they are often more suitable at carrying significant loads making them ideal for agricultural and defence applications. In extension to this, using it for surveillance of poachers, heavy cameras should not negatively impact the drone's functioning.

XI. The design

The aircraft design was sourced from Eclipson and fabricated using 3D printing, with electronic components provided by the Durham Engineering Department. The final assembly features a twin-motor configuration—one motor mounted on each wing—enabling vertical takeoff in a bi-copter arrangement. Passive stabilization during vertical flight is supported by the flaps. Upon reaching a suitable altitude, the pilot initiates the transition to forward flight by activating FBWA (Fly-By-Wire A) mode, which commands the aircraft to pitch forward significantly, facilitating the shift to horizontal flight. In this mode, the aircraft functions similarly to a conventional fixed-wing RC model, although it lacks a rudder. Yaw control is achieved via differential thrust between the two motors, while roll control is provided by elevons located on each wing.

During takeoff, the aircraft operates in Stability Mode, which enables automatic self-leveling as long as the throttle exceeds 75% of the full throttle, with the flaps and motors working in tandem to maintain balance. Additionally, a Manual Mode is available, granting the pilot full control authority over all flight surfaces and thrust.

Using PLA, the total cost for printing all parts is £10.5. In the case of a crash landing, the most common break is between one of the wings and the fuselage due to the small contact area between the two. Despite its modular design, the use of superglue mean the entire wing needed to be reprinted resulting in an additional cost of £1.8 per wing.

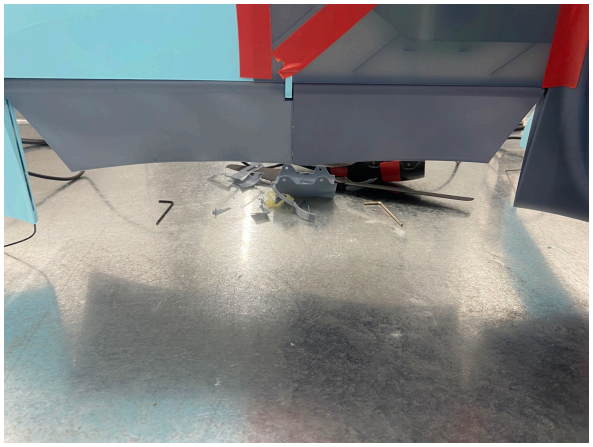


Fig. 12: Elevons

Component	Quantity	Cost per unit (£)
Holybro PM06 V2-14S Micro Power Module	1	£19.98
Holybro Pixhawk 6C Autopilot Kit PX4 Flight Controller	1	£99.72
GPS	1	£43.99
Receiver	1	£38.99
Telemetry radio	1	£56.00
AIR 20A 3-4S Multi-Rotor UAV Drone ESC	2	£71.40
AIR 2216 motor	2	£18.79
Power splitter	1	£5.00
UBEC DC-DC buck converter	1	£10.00
EMAX Servo	2	£7.99
Carbon rod	1	£5.99
D1.2 rod	4	£1.25
2300Ah battery	1	£20.75
PLA	–	£10.50

TABLE I: Bill of materials for VTOL UAV system with estimated costs

XII. Critical Review of the Design

A. *Structure*

Several critical design flaws were identified in the current iteration of the eVTOL drone, each of which significantly undermines the structural integrity, reliability, and operational robustness.

The wing sections were joined with minimal overlap and only bonded using adhesive, without any mechanical reinforcement that can be seen in the fuselage. This approach offers insufficient structural continuity, especially given the stresses seen during flight being concentrated upon the fuselage-wing joint. The absence of internal spars or a carbon fibre rod through the whole wingspan, as there is in the tail, results in a high risk of wing detachment or fatigue failure, particularly during repeated use or under moderate loading conditions.

Overall, the design lacks distributed reinforcement in high-stress areas. The reliance on being lightweight resulted in parts printed with a single wall loop, making them flimsy and easy to break. During minimal handling of the drone there was a high risk of parts breaking, with the thin walls it was easy to split apart layers with a light touch. More thought on the joints would have been useful in the design. The addition of spars in the wing-fuselage joint would improve the design.

The landing legs are inadequately designed, lacking both rigidity and appropriate clearing. Their thin profile made them prone to bending, especially after extended use as shown in figure 5, which caused the drone to be off centre upon take off. Furthermore, given their profile, there was a tendency for the legs to catch on either the fabric landing pad or the grass, causing the drone to tip and damage the wing in the first test. In the final test this problem was solved by taking off a metal surface. The design could be improved by implementing a more rigid type of leg to provide a firm base for take off, and prevent the legs getting caught.

Given the limited space within the fuselage to contain all hardware components, there were issues both in the weight distribution of the VTOL, and with magnetic interference. The hardware provided differed from the hardware that the eVTOL was designed for. This included separate flight controller and power board, which meant that the flight controller was difficult to mount in the correct orientation, leading to the manual correction for accelerometer's upright position. The battery was also a problem, with a larger battery being provided. This resulted in even less space being available within the fuselage, as well as increasing the mass of the eVTOL as a whole. With the mass being more concentrated in the battery, take-off was greatly affected. At this stage the elevons have little affect to adjust for any imbalances, leading to failure.

Due to the constricted compartment, the high-

current carrying wires were positioned very close to the flight controller and GPS, as seen in figure 4, which is likely to have caused errors with the accelerometer calibration. Using a smaller battery and combined flight controller and power board, as designed, would have improved problems with the accelerometer as the flight controller could have been fixed to the mount to be perfectly aligned. This would have also created more room to separate the high-current wires and flight controller.

B. Process

Due to the flaws in the design of the drone, multiple issues were encountered in the process of making the eVTOL. The structural flaws in the design caused parts to break easily. In parallel with the use of adhesive to bond the modular parts, this resulted in the need to re-print a large number of parts, such as a whole wing, when only one break occurred, as it was difficult to separate the bonded parts without breaking due to the single wall loop in printing. With the limited use of printers, the time needed to re-print new parts is too long for the strict time constraints of the project. This resulted in limited tests to adjust parameters in the eVTOL.

Had the drone been in a position to test the transition from vertical to horizontal flight, further logistical issues would have been faced due to the requirement to fly the eVTOL at an airstrip, a 20-minute drive away. This would have vastly increased the time to test each iteration, given that a failed one would have resulted in the need to reprint multiple parts.

During the printing itself, several issues were encountered. Due to the specified low infill density, there was trouble in the successful adhesion of the first layer onto the printing bed. This led to multiple reprints of parts with already limited time. With the thin wall loops, surface rippling was encountered on essential parts such as the elevons, which caused problems in both the assembly, and the careful handling of the eVTOL.

XIII. Other possible designs

In terms of overall design concepts, there are three primary types of small-scale eVTOL aircraft commonly developed. The first category includes tilt-rotor configurations, such as the V-22 Osprey. While this design offers both vertical take-off and efficient forward flight, it introduces structural challenges: the tilt rotor mechanisms often become critical stress points, as all aerodynamic and mechanical loads are transmitted through these components.

Tail-sitter configurations, such as the one developed in this project, offer aerodynamic efficiency due to their use of fixed, forward-facing propulsion systems. This simplicity reduces mechanical complexity compared to tilt-rotor designs. However, tail-sitters are inherently less stable during hover, particularly when utilising only two motors, as control authority in low-speed or vertical flight is limited.

The final category, often referred to as conventional or standard VTOLs, is recommended for future iterations of this project should a complete redesign be considered. These systems separate the vertical lift and forward propulsion mechanisms—typically using a multi-rotor setup for take-off and landing, and a fixed-wing with a dedicated cruise motor for forward flight. Although this design sacrifices some efficiency due to the added weight and power requirements of dual motor systems, it significantly improves stability and control. In vertical mode, the aircraft operates similarly to a quadcopter, simplifying hover and low-speed manoeuvring. Once sufficient altitude is reached, the forward propulsion system assumes control, allowing the aircraft to transition smoothly to efficient fixed-wing flight.

XIV. Other Limitations and Recommendations

A. Limitations

As well as the fundamental limitations discussed in the previous section the following factors would have impacted the ability to fully evaluate the design had it have been flyable:

- **Structural Integrity:** Although useful for reducing weight, the LW-PLA lacked in mechanical robustness. Components cracked under repeated loading and often warped soon after removal from the printer. Testing variability was also introduced due to constant repairs. For commercial use, the design of the drone would have to be far more robust ensuring repairs were less frequent.

- **Environmental Sensitivity:** Testing was heavily impacted by the weather conditions, especially the wind. Significant disturbance would be introduced during all three phases of flight. This variance and lack of repeatability made it very challenging to isolate the effects of design and tuning alterations. Furthermore, there were multiple days where the drone was ready to test but was completely grounded due to external conditions. This meant that the number of flights was far lower than intended for a complete evaluation of the drone. A controlled testing environment would be required to fully test the drone.
- **Limited Autonomy:** The operating environment of testing was clear of obstacles and predictable. This removed the need for terrain sensing and onboard obstacle avoidance. However in a complex, cluttered and unpredictable environment like a nature reserve these features would provide an additional fail safe allowing for a higher degree of autonomy. Moreover, the battery monitoring system was basic, relying on voltage thresholds alone as opposed to more useful, predictive energy models.
- **No Imaging Systems:** Although an attachment for a camera was present on the prototype, the model did not feature any of the visual or thermal imaging systems that would be essential for real world application. This limited the ability to review footage for stability and clarity.

B. Future Development to Fulfil the Problem Statement:

To enhance the system and prepare it for real-world application in environmental conservation efforts the following design improvements are recommended:

- **Material and Structural Improvements:** Commerical designs should consider the use of a more durable but still light weight material like carbon-fibre composites. Areas of critical stress should be analysed and altered to improve resistance to fatigue failure. The slightly heavier aircraft would be more resistant to small gusts as well. The inclusion of rods connecting the wing structure to the fuselage would significantly improve its structural integrity as the main weak point of the current design is the connection between the wings and fuselage.
- **Sensor Expansion and Integration of Imaging Systems:** The addition of LiDAR, optical flow and redundant barometric sensors would increase the autonomy of the platform and decrease the risk to safety of the drone and the environment it's operating in. Coupled with visual and thermal imaging features like terrain or entity following could be incorporated. Georeferenced data collected by the sensors on the aircraft is critical for habitat surveillance, mapping and anti-poaching patrols.
- **Battery and Power Management:** Smart battery systems with current monitoring, thermal management would enable more efficient energy budgeting and improve the accuracy and safety of mission planning.
- **Modular Airframe Design:** This would streamline assembly, aid in transportation and enable far easier and quicker repairs.

XV. Conclusion

This report investigated the use cases for eVTOL drones. The application of an eVTOL drone for use in antipoaching operations. It then covered the assembly and setup of the tail-sitting eVTOL drone. Through this assembly process, flaws relating to structural integrity were discovered in the design of the drone.

The report then focused on the testing process, showing how Mission Planner and Lua can be used to successfully implement autonomous flight. Ultimately, following the recommended modifications, it would be possible for a low-cost eVTOL drone to be deployed in antipoaching operations.

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